Swarada Adavadkar - Freelance AI Engineer



Production GenAI engineer: RAG, LLM evaluation and Agentic Systems

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Germany

Status: Registered freelancer in Germany, no sponsorship required

Work mode: Remote across EU/UK/Asia; on-site in Stuttgart and Baden-Württemberg (travel by arrangement)

Availability: Immediately

WHAT I DELIVER

I help teams get GenAI out of the prototype stage and into production they can trust.

- **LLM evaluation and quality audits:** build test suites and metrics for accuracy, relevance and tone; find where your assistant hallucinates and fix it.
- **RAG and GraphRAG systems:** design or rescue retrieval pipelines over internal documents, manuals and knowledge bases, including knowledge-graph retrieval with Neo4j.
- **Prototype to production:** take a GenAI proof of concept to a monitored, cost-controlled cloud service on Azure or AWS.
- **Cost and latency optimisation:** model routing, context management and retrieval tuning to cut response time and inference spend.
- **Guardrails for regulated domains:** scope restriction, refusal behaviour and safety checks for finance, insurance and industrial use cases.

Key Facts

- Experience: 5+ years in ML and AI; 2 years in production GenAI
- Industries: Fintech and financial services, insurance and telematics, industrial (energy, maritime, nuclear, defence)
- Core focus: RAG and GraphRAG, LLM evaluation, guardrails, MLOps, cloud deployment
- Cloud: Microsoft Azure, AWS
- Languages: English (fluent); German (A1, learning)
- Education: MSc Artificial Intelligence, University of Aberdeen (first class with commendation)
- Day rate: On request

PROJECT HISTORY

GenAI financial guidance assistant

2024 – 2026
UK | Remote

Client: Money Means (UK fintech)
Role: Lead AI Engineer (founding AI engineer)
Team: 2, growing to 4 AI engineers
Goal: Add a grounded AI chat assistant to an existing web platform, answering only from expert-approved financial guidance.

- Designed and shipped the RAG system from scratch, from prototype to production in **four months**.
- Built retrieval over **~350 documents** written by a chartered financial planner, plus current figures from the financial regulator; strict in-scope answering and refusal of unapproved financial advice.
- Built a test-driven evaluation suite (~50 cases per metric: factual accuracy, relevance, tone, style); improved factual accuracy by **30%** through chunking, metadata tagging and prompt design.
- **Halved response time** with model routing; moved from local prototype to monitored production service on Azure AI.
- Designed conversation flows for emotionally sensitive money conversations and different risk profiles.

Tech: Python, LangChain, LangGraph, Azure AI / Azure OpenAI, ChromaDB, RAGAS, DeepEval, FastAPI

Crash detection for usage-based insurance

2023 – 2024
UK | Hybrid

Client: The Floow (InsurTech telematics)

Role: Data Scientist
Goal: Improve crash detection from smartphone sensor data for tier-1 insurance carriers.

- Raised production crash-detection accuracy by **15%** across **500,000+** journeys with new behavioural features and model tuning.
- Built distributed pipelines with Ray to cut experimentation and training time on large sensor datasets.
- Set up MLOps with MLflow and DVC for reproducibility, dataset versioning and model governance.

Tech: Python, Ray, MLflow, DVC, scikit-learn, SQL

Predictive asset optimisation and industrial AI 2020 – 2023
Client: James Fisher AIS (digital twin software, energy/maritime/nuclear/defence) UK | Hybrid
Role: Data Science Specialist
Goal: Turn IoT and asset data into operational savings and faster knowledge discovery for industrial operators.

- Forecast machinery idling from IoT sensor data with **~80% accuracy**, contributing to an estimated **14% cut** in client operating costs.
- Designed a Neo4j knowledge graph integrating heterogeneous business data for faster semantic retrieval.
- Built computer vision pipelines on Azure ML for automated asset localisation and inspection.
- Led a proof of concept on monocular depth estimation as a lower-cost alternative to photogrammetry.

Tech: Python, Neo4j, Azure ML, PyTorch, computer vision, time-series forecasting

EDUCATION

MSc in Artificial Intelligence, *University of Aberdeen* Aberdeen, UK

BEng in Computer Science, *Savitribai Phule Pune University* Pune, India

TECHNOLOGY STACK

GenAI: RAG, GraphRAG, Agentic workflows, Tool calling, MCP, LangChain, LangGraph, LlamaIndex
Evaluation: LLM evaluation frameworks, Hallucination mitigation, Guardrails, Promptfoo, RAGAS, LangSmith, DeepEval, TruLens
Machine Learning and Deep Learning: Time-series forecasting, Computer Vision, Predictive Modelling, Feature Engineering, Model Optimisation
MLOps and Cloud: Docker, MLFlow, DVC, Ray, Azure (Azure AI, Azure ML, Functions), AWS (Bedrock, SageMaker)
Data & Retrieval: Neo4j(Graph Database), PostgreSQL/pgvector, Pinecone, FAISS
Languages & APIs: Python (expert), SQL, TypeScript, FastAPI, Pydantic, REST, GraphQL

LANGUAGES

English: Fluent

German: Basic (A1, actively learning; targeting B1 by 2027)